Kshitiz Chaurasia

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Education

Arizona State University, Tempe, AZ — *Aug. 2023 – May. 2027*
Bachelor of Science in Computer Science — GPA: 3.91/4
Relevant Coursework: Database Management, Probability & Statistics, Social Media Mining, Software QA & Testing

Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, NoSQL
Frameworks: React, Node.js, TensorFlow, PyTorch, Scikit-learn
Cloud: AWS, Azure, GCP, Docker, Kubernetes
Tools: Git, Github Actions, Postman, JIRA

Experience

NucleusTeq — *Phoenix, AZ*
AI Engineer Intern — *June 2026 – August 2026*

- Designed and built a 5-part tool-output compression engine with a 3-stage correctness check (byte-identical file paths, error codes, and version numbers) for a production LLM proxy, saving **$4,740** in LLM spend during one month of live use.
- Engineered an async, Redis-backed model router (Thompson Sampling bandit) across 6 provider endpoints with **<2ms** overhead per request, isolating genuine model-quality failures from transient errors and adding cache-state-aware cost logic to block routing decisions that would silently destroy a provider's warm prompt cache.

Lumity Media — *Tempe, AZ*
Software Developer Intern — *Jan. 2026 – May 2026*

- Architected an API integration layer (9 domain-specific client modules and a defensive normalization layer) reconciling an inconsistently-shaped, iOS-first backend across a **39.8k-line** React 19/TypeScript codebase, while keeping strict-mode TypeScript compiling clean across every surface.
- Implemented a unified mutation-state architecture using React Context to synchronize post edits, deletes, and live updates across **6 core surfaces** (Home, Explore, Library, Profile, Messages, Notifications) instantly, and fixed duplicate network calls under React StrictMode by deduplicating concurrent in-flight GET requests at the fetch layer.

Sherlocks.ai — *Palo Alto, CA*
Software Engineer Intern — *May 2025 – Aug. 2025*

- Redesigned a sequential LLM agent tool-calling pipeline into concurrent async execution (Python/FastAPI, asyncio.gather), cutting p95 tool-execution latency from **∼4.2s to ∼1.1s**.
- Diagnosed a dashboard slowdown traced to database connection-pool exhaustion from unclosed sessions, beyond the initial indexing hypothesis, then added Pydantic request validation and composite indexes on (tenant_id, created_at) for execution-log queries.

Projects

Agent Monkey: Distributed Fault-Injection Platform — *March 2026 – April 2026*

- Created a 4-service fault-injection platform (Express, Next.js, evaluation worker, MCP proxy) that intercepts tool calls across HTTP, stdio, and SSE transports to simulate **9 failure scenarios** (timeouts, rate limits, schema corruption) transparently, without modifying the agents under test.
- Built a concurrent upstream MCP connection manager (Promise.allSettled) aggregating **15+** external server tools into a unified registry, backed by a 7-migration Supabase Postgres schema with row-level security and real-time WebSocket trace streaming.

Noriv.ai: Incident Orchestration & Response Engine — *Jan. 2026 – Feb. 2026*

- Developed a multi-tenant incident response backend spanning **20+** REST/WebSocket endpoints with object-level authorization and a Redis-backed token-bucket rate limiter enforcing per-tenant request caps with 429 handling.
- Built a versioned runbook DAG execution engine with append-only audit logging for privileged state transitions, executing multi-step task graphs with bi-directional sync to Slack and Jira.

Leadership

- **Project Lead, DevLabs:** Led development of a FastAPI + PostgreSQL microservice with REST APIs and RBAC, shipping **6** releases across 4 sprints.